SIYANG LI

Homepage ⋄ Google Scholar ⋄ lsyyoungll@gmail.com

Education

Huazhong University of Science and Technology, China **2022 – 2026**
Ph.D. in Artificial Intelligence, School of Artificial Intelligence and Automation

Boston University, USA **2020 – 2021**
Master in Artificial Intelligence, College of Arts and Sciences

New York University, USA **2014 – 2018**
Bachelor in Mathematics and Computer Science, Courant Institute

Research Experience

Transfer Learning for Calibration-Efficient Brain-Computer Interfaces **2021 – 2026**
Advisor: **Prof. Dongrui Wu (IEEE Fellow; EiC, IEEE Trans. Fuzzy Systems)**
Huazhong University of Science and Technology, China
Brain-computer interface (BCI) is an interdisciplinary field covering biomedical engineering, computer science, etc. **EEG**-based non-invasive BCIs face challenges of individual variability and signal non-stationarity, making per-use calibration necessary yet inconvenient. My research develops **transfer learning** methods that align auxiliary data from other subjects, devices, or modalities with the target user, thereby reducing reliance on target-specific data. This enables fast and accurate **BCI decoding** with improved robustness and stronger privacy protection.

Language Modeling for Low-Resource Machine Translation **2020 – 2021**
Advisor: **Prof. Derry Wijaya**
Boston University, USA
Large language models (LLMs) have transformed natural language processing, but internet text corpora are dominated by high-resource languages such as English and Mandarin. My research developed **transfer learning** approaches to adapt pre-trained LLMs on mainstream languages to low-resource languages, enabling efficient transfer with minimal aligned corpora. This enables improved **machine translation** quality for data-scarce languages and scenarios with limited data.

Research Interests

My research interests include **machine learning** and its applications. Technically, I study **transfer learning**, **ensemble learning**, and **multimodal learning** methods and theories. I am also familiar with modern **large language models and vision-language models (LLMs/VLMs)**. For applications, I am particularly interested in brain decoding for **brain-computer interfaces (BCIs)**. In the long term, I aim to build **multimodal human-computer interaction** systems for future generations.

Publications

Machine Learning & AI

[1] **S. Li†**, C. Liu†, D. Wu*, Z. Zeng*, and L. Ding*, "StackingNet: Collective Inference Across Independent AI Foundation Models," **Advanced Science**, e76488, 2026. URL. Code.
Aggregates only the output predictions of independently developed black-box foundation models at inference, with no access to weights or training data, beating every individual model and classic ensembles on language, vision and paper-rating tasks, while also ranking model reliability and pruning models that drag the group down.

[2] **S. Li**, Z. Wang, C. Liu, and D. Wu*, "Black-Box Test-Time Ensemble," IEEE Computational Intelligence Magazine, vol. 21, no. 1, pp. 57-68, 2026. URL. Code.
Extends the spectral meta-learner from binary to multi-class so the accuracy of API-only models can be estimated from their predictions on unlabeled test data alone; the resulting hyperparameter-free weighting updates in milliseconds per sample and beat majority voting on 13 text, image and time-series datasets.

[3] Z. Cui, **S. Li**, S. Kan, Y. Wang, F. Luo, D. Wu*, "A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces," IEEE Transactions on Emerging Topics in Computational Intelligence, early access, 2026. URL.
Replaces crisp classes with fuzzy ones to approximate class-conditional probabilities for continuous labels, so classification-based transfer learning algorithms extend to regression with a derived target-risk bound, validated on three EEG regression datasets including driver drowsiness estimation.

[4] H. Wang, H. Zhang, **S. Li**, D. Wu*, "Gated Parametric Neuron for Spike-based Audio Recognition," Neurocomputing, vol. 609, pp. 128477, 2024. URL. Code.
Adds gates to the leaky integrate-and-fire neuron so gradients survive the membrane-potential path, and learns the time constant and firing threshold instead of hand-setting them, reproducing the spatio-temporal heterogeneity of real neurons and improving spike-based audio recognition.

[5] G. Kuwanto†, A. F. Akyürek†, I. C. Tourni†, **S. Li†**, A. Jones, and D. Wijaya*, "Low-Resource Machine Translation Training Curriculum Fit for Low-Resource Languages," in Proc. Pacific Rim International Conference on Artificial Intelligence, 2023. [co-first author] URL.
Pre-trains on code-switched text and Wikipedia sentences mined with an induced dictionary rather than parallel corpora, under compute budgets scaled to local incomes; the curriculum beats supervised translation on the same backbone for Gujarati, Somali and Kazakh.

[6] H. Wu, **S. Li**, and D. Wu*, "TMMM: Transformer in Multimodal Sentiment Analysis under Missing Modalities," Proc. International Joint Conference on Neural Networks, 2024. URL.
Trains with modalities deliberately dropped and reconstructs the missing ones at test time, adding classification-token fusion, a mixture-of-experts block and contrastive pre-training, improving over the best baseline by about 2% and 2.5% on two sentiment datasets.

[7] **S. Li**, Y. Xu, H. Wu, D. Wu*, Y. Yin, J. Cao, and J. Ding, "Facial Expression Recognition In-the-Wild with Deep Pre-trained Models," in Proc. European Conf. on Computer Vision Workshop, 2022. URL. Code.
Names four failure modes of in-the-wild expression recognition —uncertain labels, class imbalance, pre-training/downstream mismatch, and single-architecture bias— and supplies one module for each, transferring knowledge from synthetic faces to real ones in the ABAW4 challenge.

Accurate Brain–Computer Interface Decoding

[1] **S. Li**, Z. Wang, H. Luo, L. Ding*, and D. Wu*, "T-TIME: Test-Time Information Maximization Ensemble for Plug-and-Play BCIs," IEEE Transactions on Biomedical Engineering, vol. 71, no. 2, pp. 423-432, 2024. URL. Code.
Adapts an ensemble of classifiers to each arriving unlabeled EEG trial by conditional entropy minimization and adaptive marginal distribution regularization, so a new user needs no calibration session at all; the first test-time adaptation work for EEG, outperforming about 20 transfer learning approaches.

[2] **S. Li†**, J. Ouyang†, Z. Cui†, Z. Wang, T. Jia, F. Wan, and D. Wu*, "Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces," IEEE Journal of Biomedical and Health Informatics, early access, 2026. URL. Code.
Replaces gradient-based test-time adaptation with forward-pass-only transformations of each trial, weighted by a learning-to-rank module, cutting computation and keeping model weights private; verified on three motor imagery and two driver drowsiness datasets, including under simulated electrode and movement noise.

[3] **S. Li**, Z. Wang, S. Zheng, and D. Wu*, "Heterogeneous Supervised Domain Adaptation (HSDA) for Cross-Device EEG Classification," IEEE Transactions on Pattern Analysis and Machine Intelligence, R3, under review. Code.
Maps EEG from headsets with different electrode counts and locations into a shared latent space through device-specific transformations, then adds prototype learning and model-inversion gradient regularization, raising accuracy by 7.19% on motor imagery and AUC by 5.23% on P300 over same-device transfer.

[4] **S. Li**, H. Wu, L. Ding, and D. Wu*, "Meta-Learning for Fast and Privacy-Preserving Source Knowledge Transfer of EEG-Based BCIs," IEEE Computational Intelligence Magazine, vol. 17, no. 4, pp. 16–26, 2022. URL. Code.
Learns an initialization across many previous users so a new one is decoded from a handful of labeled trials, sharing only model parameters and never raw EEG; the first approach to address individual differences, data scarcity and privacy simultaneously.

[5] **S. Li**, H. Wang, X. Chen, and D. Wu*, "Multimodal Brain-Computer Interfaces: AI-powered Decoding Methodologies," arXiv preprint, 2025. arXiv.
Organizes multimodal brain decoding into fusion, instance-wise cross-modality mapping and sequential translation across visual, speech and affective applications, and documents pitfalls such as block-design temporal leakage that silently inflate reported accuracy.

[6] Z. Wang†, **S. Li†**, J. Luo, J. Liu, D. Wu*, "Channel Reflection: Knowledge-Driven Data Augmentation for EEG-based Brain-Computer Interfaces," Neural Networks, vol. 176, pp. 106351, 2024. [co-first author] URL. Code.
Exploits the left-right symmetry of the scalp: mirroring channels across the midline and swapping the label yields physiologically valid new trials at zero parameter cost, improving accuracy on eight datasets spanning motor imagery, SSVEP, P300 and seizure detection.

[7] H. Wu, **S. Li**, and D. Wu*, "Motor Imagery Classification for Asynchronous EEG-Based Brain-Computer Interfaces," IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 32, pp. 527-536, 2024. URL. Code.
Removes the need for trial-start triggers by first screening each sliding window for rest versus imagery and only then

classifying the task, with self-supervised refinement of both feature extractors beating the best baseline by about 2% on four datasets.

[8] X. Chen, **S. Li**, D. Wu*, "AFPM: Alignment-based Frame Patch Modeling for Cross-Dataset EEG Decoding," Science China Information Sciences, vol. 69, no. 6, 162403, 2026. URL. Code.
Selects channels by brain-region priors, Euclidean-aligns distributions and remaps everything onto a common layout so one transformer can be pre-trained across fourteen datasets; with no target-domain calibration it gains up to 4.40% on motor imagery and 2.98% on ERP over seventeen competing models.

[9] D. Liu, **S. Li**, Z. Wang, W. Li, D. Wu*, "Spatial Distillation based Distribution Alignment (SDDA) for Cross-Headset EEG Classification," IEEE Transactions on Biomedical Engineering, vol. 73, no. 7, pp. 2400-2411, 2026. URL. Code.
Instead of cropping to the electrodes two headsets share, distills the extra channels' spatial knowledge into the smaller montage and then aligns input, feature and output spaces, beating ten transfer learning approaches on six datasets in both offline and online calibration.

[10] Z. Wang, **S. Li**, X. Chen, and D. Wu*, "Time-Frequency Transform based EEG Data Augmentation for Brain-Computer Interfaces," Knowledge-Based Systems, vol. 311, pp. 113074, 2025. URL. Code.
Decomposes trials by wavelet or Hilbert-Huang transform, reassembles the sub-signals across different subjects, and reconstructs them in time, so one operation attacks data scarcity, non-stationarity and individual differences together, gaining about 4% over baselines across seventeen datasets.

[11] Z. Wang, **S. Li**, X. Chen, and D. Wu*, "MVCNet: Multi-View Contrastive Network for Motor Imagery Classification," Knowledge-Based Systems, vol. 328, pp. 114205, 2025. URL. Code.
Runs a convolutional and a transformer branch in parallel rather than in series, tied by two contrastive losses —one across augmented time, frequency and spatial views, one across the two branches— outperforming nine state-of-the-art motor imagery networks on five datasets.

[12] Z. Wang, H. Wang, T. Jia, X. He, **S. Li**, D. Wu*, "DBConformer: Dual-Branch Convolutional Transformer for EEG Decoding," IEEE Journal of Biomedical and Health Informatics, vol. 30, no. 5, pp. 4134-4147, 2026. URL. Code.
Splits decoding into a temporal branch for long-range dependencies and a spatial branch for inter-channel relations, with channel attention weighting electrodes; it beats twelve baselines on motor imagery, seizure detection and SSVEP using about one-eighth the parameters of EEG Conformer, with filters matching sensorimotor priors.

[13] X. Chen, J. An, H. Wu, **S. Li**, B. Liu, and D. Wu*, "Front-End Replication Dynamic Window (FRDW) for Online Motor Imagery Classification," IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 31, pp. 3906–3914, 2023. URL. Code.
Classifies before a trial has finished by repeating the partial signal up to the training length and committing as soon as confidence suffices, raising the information transfer rate; the approach helped win the national championship of the 2022 China BCI Competition.

[14] C. Liu, **S. Li**, L. Tan, and D. Wu*, "Temporal Out-of-Distribution Detection for Asynchronous Motor Imagery Brain-Computer Interfaces," arXiv preprint, 2026. arXiv. Code.
Reframes continuous-stream control as rest/task gating followed by joint classification and rejection, scoring novelty with output energy, deep-feature density and short-term temporal consistency, so unmodeled mental states are rejected rather than turned into spurious commands.

Clinical & Trustworthy Brain Decoding

[1] Z. Wang, **S. Li**, D. Wu*, "Canine EEG Helps Human: Cross-Species and Cross-Modality Epileptic Seizure Detection via Multi-Space Alignment," National Science Review, vol. 12, no. 6, nwaf086, 2025. URL. Code.
Aligns canine with human EEG and scalp with intracranial recordings in the input, feature and output spaces via domain adaptation and knowledge distillation, exceeding 90% AUC for seizure detection with only about 10% of the target species labeled —evidence that animal data can supply the scale human EEG lacks.

[2] **S. Li†**, Z. Wang†, X. Gui†, X. Chen, Z. Wang, Y. Wen, and D. Wu*, "RAICL: Retrieval-Augmented In-Context Learning for Vision-Language-Model Based EEG Seizure Detection," IEEE Transactions on Systems, Man and Cybernetics: Systems, R1, under review, preprint available: arXiv. Code.
Renders multichannel EEG as stacked waveform plots and has an off-the-shelf vision-language model read them, retrieving the most relevant labeled plots as in-context examples; without any retraining it matches signal-trained seizure detectors, with retrieval adding over 10% balanced accuracy over random example selection.

[3] Z. Wang, W. Zhang, **S. Li**, X. Chen, and D. Wu*, "Unsupervised Domain Adaptation for Cross-Patient Seizure Classification," Journal of Neural Engineering, vol. 20, no. 6, pp. 066002, 2023. URL. Code.
Screens out dissimilar source patients, globally aligns the remaining ones with the new patient, then borrows labels from the single closest patient, beating thirteen approaches on two datasets and raising sensitivity by about 10%, so clinicians need not label each new patient.

[4] H. Wang, Z. Jia, Y. Shen, Z. Wang, **S. Li**, Z. Tang, K. Shu, F. Hu, D. Wu*, "SACM: SEEG-Audio Contrastive Matching for Chinese Speech Decoding," IEEE Transactions on Biomedical Engineering, early access, 2026.

URL. Code.
Introduces a Mandarin speech-decoding protocol with stereo-EEG and synchronized audio from ten epilepsy patients and matches neural activity to the spoken audio contrastively, raising top-1 accuracy by 40.7% over signal-only baselines, with a single ventral sensorimotor cortex electrode nearly matching the full array.

[5] Y. Tu, **S. Li**, X. Chen, D. Wu*, "BrainprintNet: A Multiscale Cross-Band Fusion Network for EEG-based Brainprint Recognition," IEEE Transactions on Information Forensics and Security, vol. 21, pp. 2757-2768, 2026. URL. Code.
Combines fine-grained filter banks, grouped multiscale temporal convolutions and cross-band spatial fusion to identify people from EEG when both session and task change —the realistic condition earlier work avoided— improving cross-session accuracy by 5.35% and revealing which bands and channels carry identity.

[6] X. Chen, **S. Li**, Y. Tu, Z. Wang, D. Wu*, "User-wise Perturbations for User Identity Protection in EEG-Based BCIs," Journal of Neural Engineering, vol. 22, no. 1, pp. 016040, 2025. URL. Code.
Adds one perturbation per user —random, synthetic, error-minimizing or error-maximizing— that renders identity unlearnable from released EEG while the task label stays fully decodable, verified on six datasets against both deep and traditional identification models.

[7] T. Jia, L. Meng, **S. Li**, D. Wu*, "Federated Motor Imagery Classification for Privacy-Preserving Brain-Computer Interfaces," IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 32, pp. 3442-3451, 2024. URL. Code.
Keeps batch-normalization statistics local to each user so raw EEG never leaves the device, and trains with sharpness-aware minimization for flatter minima, beating six federated learning approaches and even centralized training, so privacy here costs no accuracy.

[8] X. Jiang, L. Meng, **S. Li**, and D. Wu*, "Active Poisoning: Efficient Backdoor Attacks on Transfer Learning-based Brain-Computer Interfaces," Science China Information Sciences, vol. 66, no. 8, 182402, 2023. URL.
Shows that publicly shared source EEG can be poisoned with a narrow periodic pulse so any model transfer-trained on it misclassifies triggered trials while behaving normally otherwise, with active selection of which trials to poison raising attack success from 83.2% to 94.3%.

Awards

- Hua-Nao (China-Brain) Award, Top-10 Highlights in China's Brain-Computer Interface, *Transfer Learning for Accurate EEG Decoding*, Contributors: Dongrui Wu, Ziwei Wang, **Siyang Li**, Tianwang Jia. 2026. News
- Hua-Nao (China-Brain) Award, Top-10 Highlights in China's Brain-Computer Interface, *Knowledge-Data Fusion for Accurate and Efficient Brain-Computer Interface Decoding*, Contributors: Dongrui Wu, Xue Jiang, Ziwei Wang, Ruimin Peng, Lubin Meng, **Siyang Li**. 2024. News.
- National Scholarship (Doctoral Students), 2024.
- Youth Paper Defense Award, World Robot Contest - Brain-Computer Interface Track, 2023.

Funding

- 20,000 CNY. Grant from the Henan Key Laboratory of Brain Science and Brain-Computer Interface Technology Open Fund (HNBBL230204), *Research on Transfer Learning Algorithms for Calibration-Free Brain-Computer Interfaces*, 2023–2025. URL.

Talks

- **S. Li**, D. Wu, Tutorial: Machine Learning Pipeline for EEG-based Brain-Computer Interfaces, International Joint Conference on Neural Networks, Gold Coast, Australia, 2023.
- D. Wu, Z. Wang, **S. Li**, Tutorial: Transfer Learning for EEG-based Brain-Computer Interfaces, International Conference on Neural Information Processing, Changsha, China, 2023.

Academic Service

Reviewer for IEEE Transactions on Biomedical Engineering, IEEE Transactions on Neural Systems and Rehabilitation Engineering, IEEE Computational Intelligence Magazine, IEEE Transactions on Consumer Electronics, and related journals.

Patents

- D. Wu, **S. Li**, H. Wu. Meta-learning methods, EEG signal recognition methods, and devices for brain–computer interfaces. Chinese Invention Patent, CN114879838A, *granted*.
- D. Wu, **S. Li**, Z. Wang, S. Zheng. Construction method, classification method, and system for cross-device EEG classification models. Chinese Invention Patent, CN119719914A, *granted*.

- D. Wu, **S. Li**. Test-time adaptation methods and systems for calibration-free brain–computer interfaces. Chinese Invention Patent, CN117056837A, *granted*.
- D. Wu, **S. Li**, Z. Wang. Test-time ensemble method, medium, and device for black-box models. Chinese Invention Patent, CN118626848A, *published*.
- D. Wu, H. Wu, **S. Li**. Asynchronous motor imagery BCI recognition methods and systems. Chinese Invention Patent, CN117034105A, *granted*.
- D. Wu, X. Jiang, L. Meng, **S. Li**. Training methods and devices for machine learning models with backdoors, and electronic equipment. Chinese Invention Patent, CN114021626A, *published*.
- D. Wu, Z. Wang, **S. Li**, X. Chen. Supervised contrastive learning methods and devices based on spatio-temporal-spectral consistency. Chinese Invention Patent, CN118626901A, *published*.
- D. Wu, **S. Li**. Stroke rehabilitation methods based on the integration of motor imagery BCI and transcranial direct current stimulation. Chinese Invention Patent, CN120420602A, *published*.
- D. Wu, D. Liu, **S. Li**, Z. Wang. Construction methods and classification methods and systems for EEG models across different devices. Chinese Invention Patent, CN120067794A, *published*.